

Treat the following questions:

1. Define the following terms: (5pts)
 - a. HTTPS.
 - b. GET Method.
 - c. HEAD Method.
 - d. Intrusion Prevention System (IPS).
 - e. Remote Procedure Call (RPC).
 - f. Socket.
 - g. HTTP method idempotent.
 - h. HTTP method safe.
 - i. RMI.
 - j. InetAddress.
2. **Answer by true or false, in case of false explain why.** (5pts)
 - a. The POST method is idempotent and is not safe.
 - b. The advantage of CORBA is that it allows the system to work with objects that are written with the same language.
 - c. RMI is slower than RPC because RMI involves running Java bytecode.
 - d. RMI is procedural like in C, but RPC is object oriented.
 - e. The "STUB" is an object, acts as a gateway for the server side.
 - f. TCP is a "Full-Duplex" Protocol.
3. List and explain the functionalities of the mediator (Middleware). (2pts)
4. Talk about the advantages and disadvantages of the client/server architecture. (2pts)
5. Create in Java two programs named Server.java and Client.java that communicate with each other as specified below: (6pts)
 - The server is called hostname.
 - A socket of type ServerSocket is created on port 10452.
 - Ensure the connection between the Client and the Server.
 - Once the client is connected, the server displays on the screen the following message: « Connection Request Received From yyy », where yyy is the host name of the client.
 - The Server sends the client the message 'Server> Salam', the client displays the message on the screen.
 - The client sends the message 'Client> Ahla' to the server and at the end the server displays the received message on screen.